Satisfiability implies consistency

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub thm:soundness (this batch); note the preceding cor:weak-soundness (source lines 1823-1826) has no proof environment of its own in the source and is not a counted card - the census pairing artifact is recorded here.

Editorial note. The label cor:consistency-soundness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Soundness theorem. Same unit, \ollabel{thm:soundness} (source line 1759, this batch). Satisfying structures make provable-from-Gamma formulas true, in particular falsum is never satisfiable.

Source reference. Original theorem thm-soundness: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 1759–1759. Prerequisite pointer invoked at proof line(s) 1836; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Corollary 0.1. *If Γ is satisfiable, then it is consistent.*

Proof. We prove the contrapositive. Suppose that Γ is not consistent. Then $\Gamma \vdash \bot$, i.e., there is derivation of $\bot$ from Γ. By thm-soundness, any structure $\mathfrak{M}$that satisfies Γ must satisfy $\bot$. Since $\mathfrak{M} \nvDash \bot$ for every structure $\mathfrak{M}$, no $\mathfrak{M}$can satisfy Γ, i.e., Γ is not satisfiable. □